

Agent-Based Monte Carlo Simulation of Tumor Growth

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1 Introduction

Tumor growth emerges from the collective behavior of many interacting cells within a spatially structured microenvironment. Local effects such as cell–cell contact, spatial constraints, and limited nutrient availability play a central role in shaping tumor morphology and growth dynamics. These features are difficult to capture using continuum models, which motivates the use of agent-based modeling approaches.

Agent-based models (ABMs) describe a system in terms of individual cells that follow simple, local rules depending on their state and immediate surroundings. In cancer growth modeling, ABMs are particularly suitable for studying spatial heterogeneity and emergent behavior resulting from microscopic interactions. A common formulation employs on-lattice models, where cells occupy sites on a regular grid and interact with neighboring cells.

The stochastic nature of cellular processes such as division, death, and mutation is naturally incorporated using Monte Carlo methods, where events occur probabilistically at each time step. In this project, an on-lattice agent-based Monte Carlo model is implemented in the Julia programming language. The implementation deliberately relies on simple data structures using only statics and randoms, without the use of specialized or high-level simulation packages.

The aim of this work is to qualitatively explore tumor growth under oxygen-limited conditions and to demonstrate how simple coding choices and local interaction rules can give rise to realistic growth behavior and spatial organization.

2 Model and Methods

The tumor is modeled using an on-lattice agent-based framework implemented in the Julia programming language. Cells occupy sites on a two-dimensional square lattice of size $N \times N$, in our model we set $N=301$. Each site can be empty, contain a normal cell, a cancer cell, or a dead cell. The system is initialized with a single normal cell placed at the center of the lattice. Cell dynamics are updated using a Monte Carlo scheme: at each time step, all living cells are visited once in random order, and stochastic decisions for division or death are made based on local oxygen availability. A single mutation event is introduced by converting a normal cell at the tumor boundary into a cancer cell. the time of introducing the mutation is controled by in

the model to test the different behavior of the cancer growth in different mutation time, where we have tested at time $T_MUT=120$ and $T_mut = 60$ in our simulation. and different seeds were used to generate different realizations of the stochastic process.

Division and death probabilities differ for normal and cancer cells with cancer having a higher probability for division and a lower probability for death. The division and death rates depend explicitly on the oxygen concentration at the cell's location.

The oxygen field is prescribed as a static, radially symmetric profile that is lowest at the tumor core and increases toward the lattice boundary, mimicking diffusion-limited nutrient supply. It is defined as

$$O(r) = O_{\min} + (1 - O_{\min}) \left(1 - e^{-r/L}\right), \quad (1)$$

where r is the distance from the lattice center, $O_{\min}$ is the minimum oxygen level at the core, and L controls the characteristic length scale of oxygen recovery. Cell division is attempted first by placing a daughter cell in a neighboring empty site; if no empty site is available, division may still occur through a pushing mechanism, where a chain of neighboring cells is displaced along a lattice direction until an empty site is reached. Cancer cells are assigned a higher probability to push than normal cells. Dead cells are removed stochastically with a fixed clearance probability, allowing space to be freed over time.

3 Results and Discussion

3.1 Tumor growth dynamics

3.2 Spatial structure

3.3 Effect of oxygen limitation

4 Conclusion and Outlook